

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

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Section / Semester: G / V Semester
GitHub Repository Link: https://github.com/Saatvik-G/SE-LABS_PES1UG24CS395.git

Jira Workspace Link: <https://pes-se-labs.atlassian.net/jira/software/projects/CK>

1. Product Backlog (Epics & User Stories)

Epic Name	Key	User Story Description	Priority	SP (Est)
Epic 1: Placement & Capacity	CK-1	As a Warehouse Operator, I want to automatically validate the weight of a pallet against the rack's capacity, so that I do not overload the rack.	High	5
Epic 1: Placement & Capacity	CK-2	As a Warehouse Operator, I want to request a supervisor override when a rack capacity is exceeded, so that I can place the pallet if verified safe.	Medium	3
Epic 1: Placement & Capacity	CK-3	As a Logistics Supervisor, I want to configure the 3D layout coordinates and maximum weight capacities of racks, so that safety limits remain current.	Medium	5
Epic 2: Stock Tracking	CK-4	As a Warehouse Operator, I want stock movements to log automatically when a barcode/RFID tag is scanned, so that changes are tracked in real-time.	High	8
Epic 2: Stock Tracking	CK-5	As a Warehouse Operator, I want to search for a pallet's 3D coordinates using its SKU code, so that I can locate and retrieve it quickly.	High	3
Epic 2: Stock Tracking	CK-6	As a Logistics Supervisor, I want to receive an alert when a pallet is placed in an incorrect slot, so that I can correct inventory errors immediately.	Medium	2
Epic 2: Stock Tracking	CK-7	As a Logistics Supervisor, I want the system to generate weekly automated audit reports, so that I can monitor stock discrepancies.	Medium	5
Epic 2: Stock Tracking	CK-8	As a Logistics Supervisor, I want to view a historical log UI of all pallet location updates, so that I can audit inventory movements.	Low	3

2. Jira Backlog View Evidence

Product Backlog (Jira View)

Project: CK (Coffee-Kiosk)

Epic 1: Pallet Placement & Capacity Validation

CK-1

Story 1.1: Automatic Capacity Check

Capacity-Epic

▲

5

CK-2

Story 1.2: Supervisor Overweight Override

Capacity-Epic

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3

CK-3

Story 1.3: Configure Rack Limits

Capacity-Epic

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Epic 2: Stock Tracking & Retrieval Experience

CK-4

Story 2.1: Barcode/RFID Stock Logging

Tracking-Epic

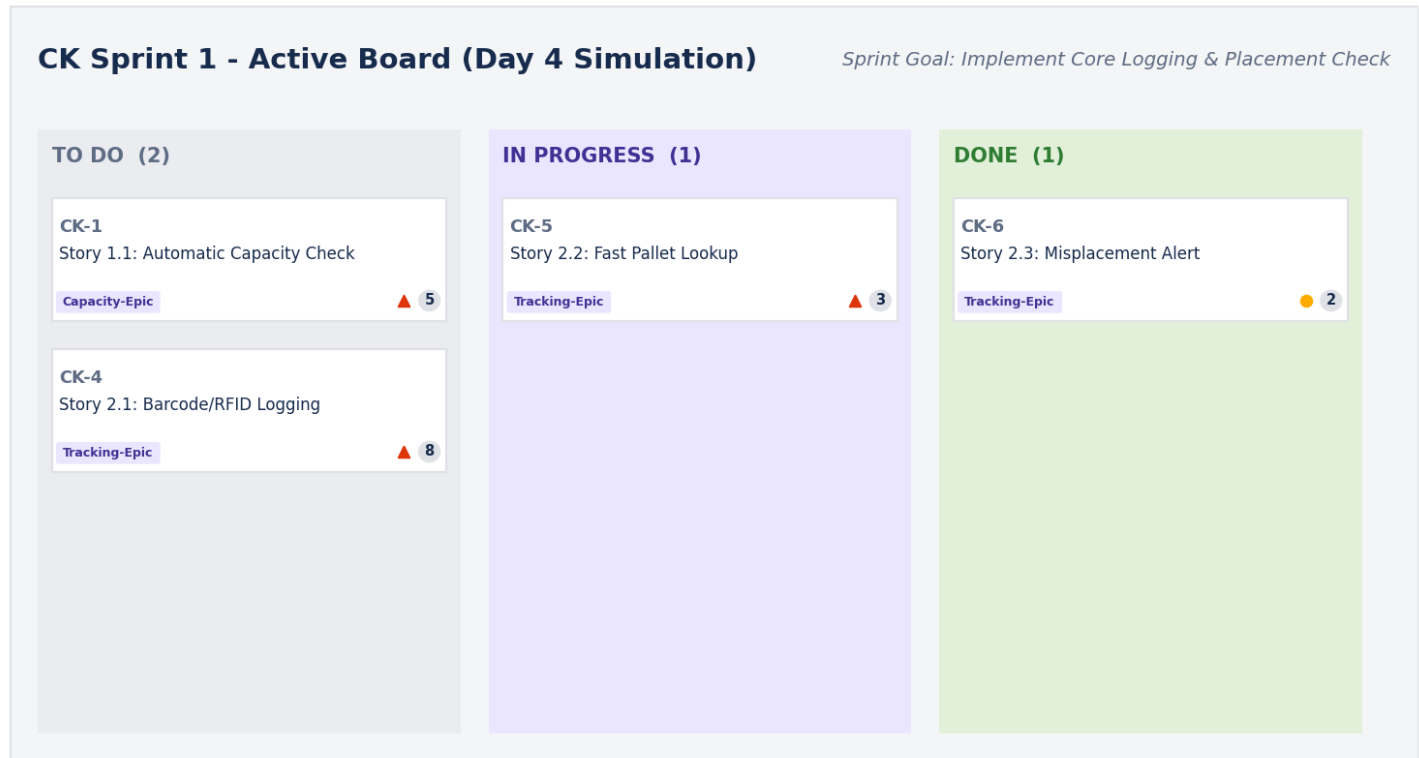
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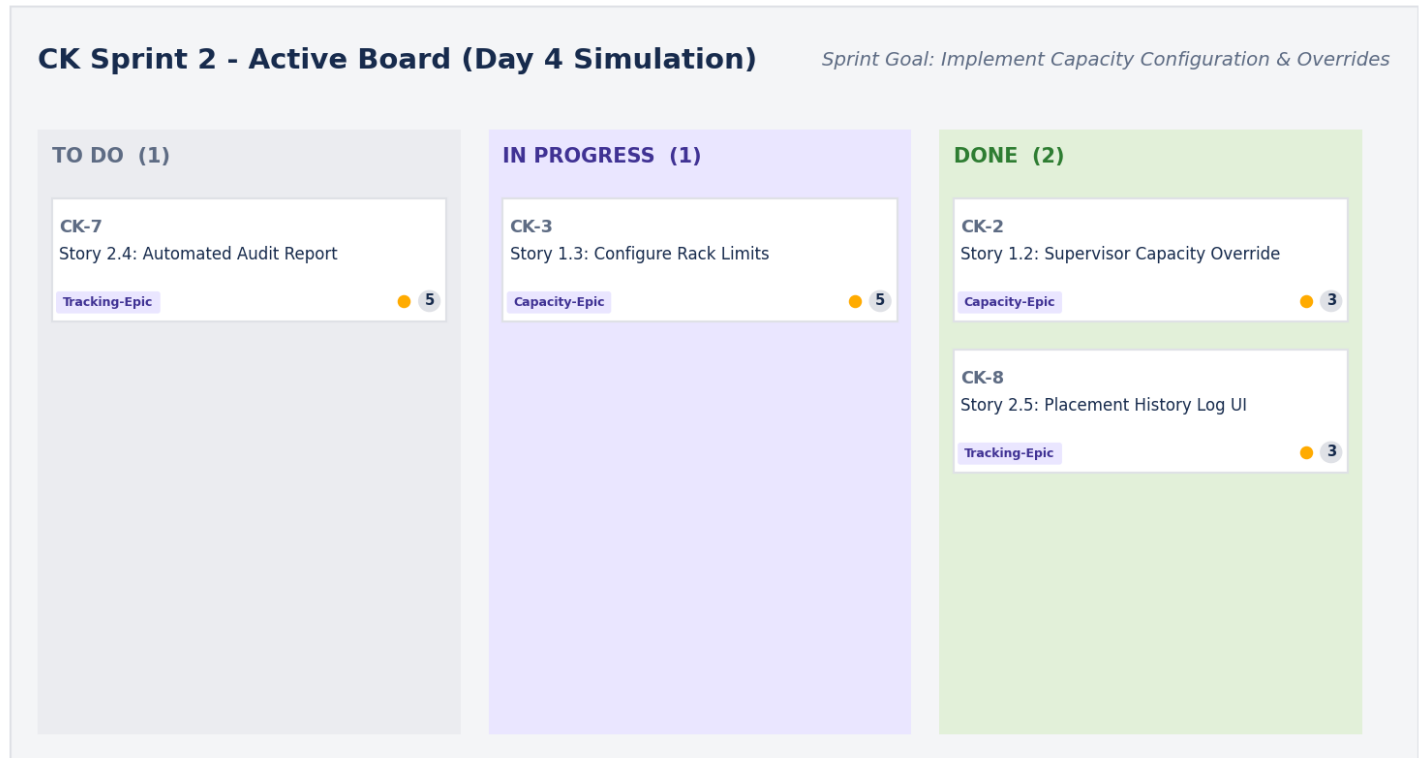
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3. Sprint Simulation Board Screenshots

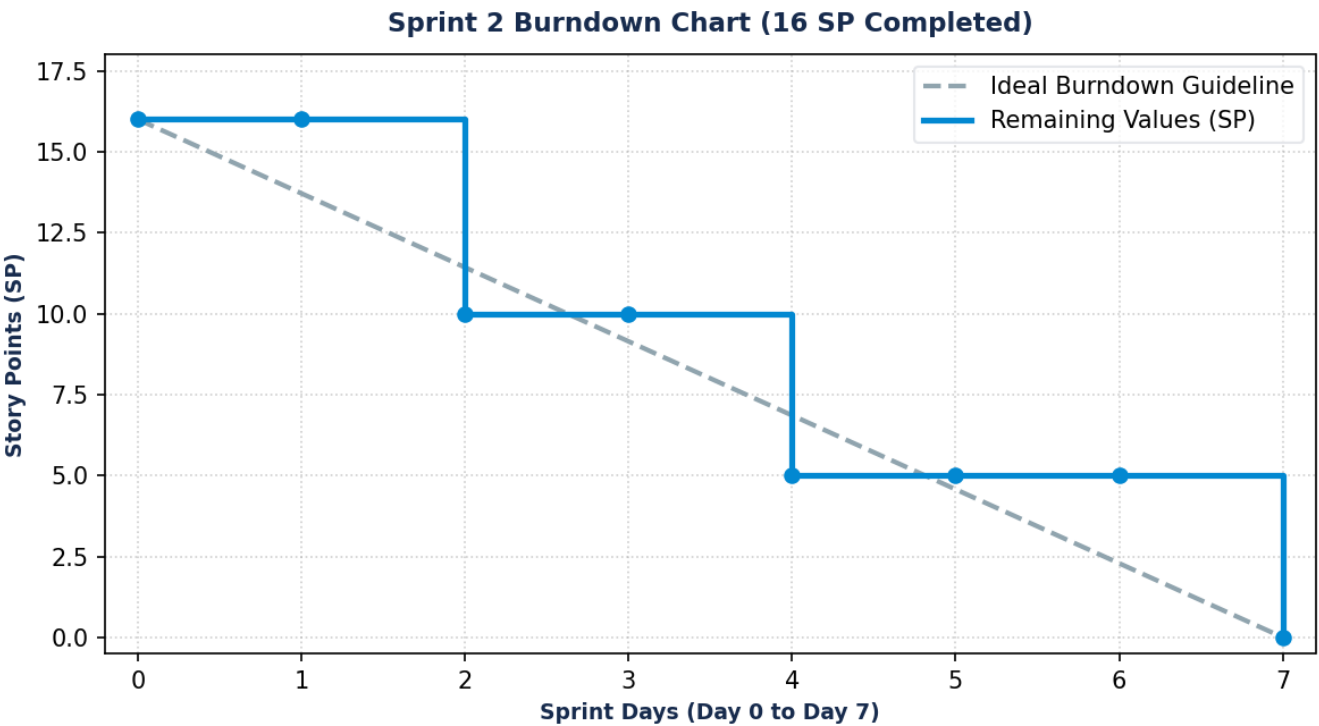
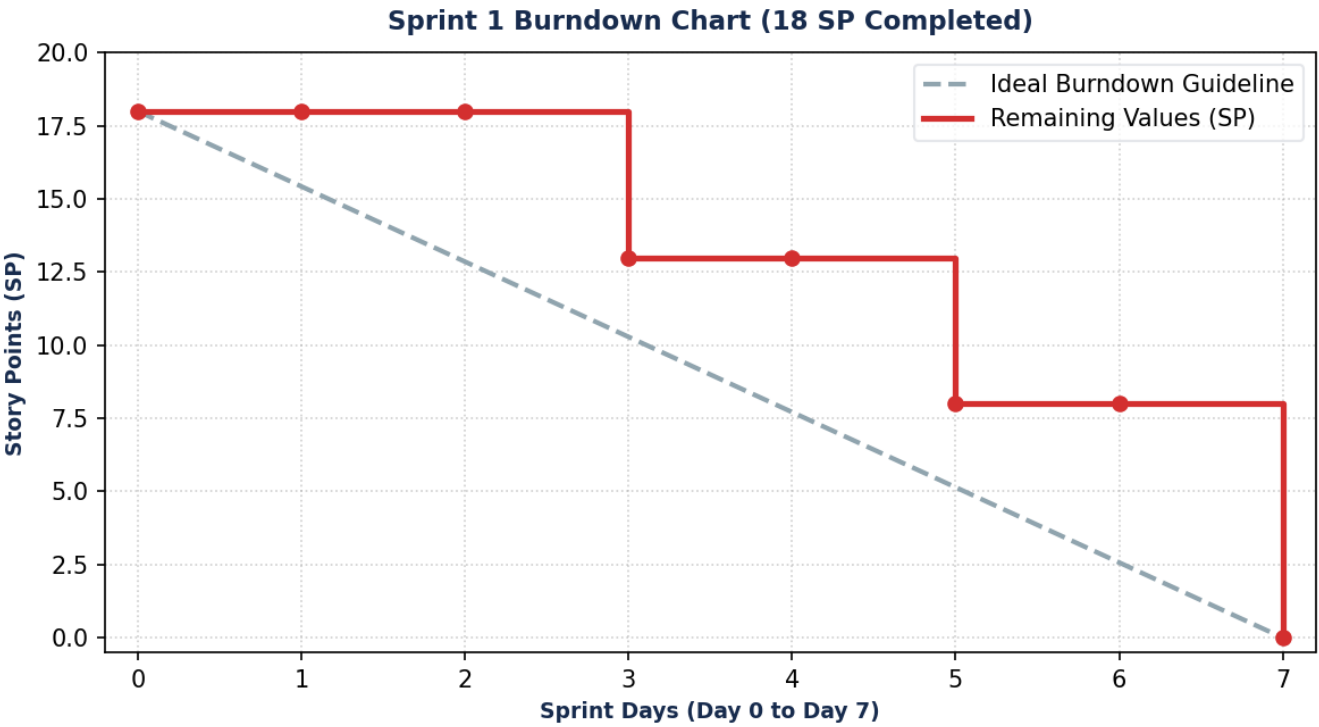
Sprint 1 Board (Day 4 Simulation) - Velocity Committed: 18 SP



Sprint 2 Board (Day 4 Simulation) - Velocity Committed: 16 SP



4. Sprint Burndown Chart Evidence



5. Agile Reflection Answers

Q1: Did your estimations reflect the actual effort?

Yes, in both sprint simulations, the relative story point estimations matched actual effort well. Story CK-4 (barcode/RFID logging, 8 SP) was estimated high due to hardware interface complexity and database synchronization; it consumed the most development time and was completed on the final day. Story CK-1 (capacity validation check, 5 SP) took a moderate amount of effort. In Sprint 2, the team similarly found that the weekly audit report generation (CK-7, 5 SP) required significant report layout logic and queries, while simple logs (CK-8, 3 SP) were fast. Story points served as an accurate index of task difficulty.

Q2: Was your backlog well-prioritized?

Yes, backlog items were prioritized based on risk, system safety, and core value. Critical safety features (CK-1, checking rack overload limits) and core system utilities (CK-4, stock scanning logging) were prioritized as High and pulled into Sprint 1. Workflow enhancements and supervisors' configurations (CK-2, overrides, and CK-3, limits editing) were prioritized as Medium and placed in the Product Backlog to be resolved during Sprint 2, ensuring that safety-critical functionality was verified and deployed first.

Q3: How did your simulated sprint align with your plan?

The team completed both sprints fully within the 1-week time-boxes. Sprint 1 completed its committed scope of 18 SP on time, and Sprint 2 achieved its 16 SP commitment. In both sprints, resolving smaller, lower-estimate tasks by Day 3 (like CK-5 and CK-6 in Sprint 1; CK-2 and CK-8 in Sprint 2) established early momentum and provided a buffer for more complex, high-point stories (like the 8 SP logging in Sprint 1 and 5 SP report in Sprint 2) to be completed by the final day.

Q4: What insights did the burndown chart give about your team's capacity?

Both burndown charts exhibit a stepped pattern (flat lines followed by vertical drops) rather than a smooth linear slope. This reflects the nature of our tasks: several stories were estimated at 5 SP or 8 SP, which meant points were only 'burned' at the end of the development cycle once the entire task was resolved. Insight: An 8 SP task (like CK-4) is too large for a 1-week sprint because it limits visibility. In future planning sessions, we will break down any user stories estimated above 5 SP into smaller sub-tasks (e.g., separating the DB logging from the hardware integration) to facilitate a smoother, more trackable burn curve.